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Static eliminator

Abstract

To suppress charging of a housing of a static eliminator while avoiding an influence on control of ion balance. In order to suppress the charging of the housing of the static eliminator, the housing is provided with a rear frame that is conductive. The rear frame is insulated from a placement surface of the static eliminator by an insulating member such as an insulating pad, an inner spacer, and an outer spacer, and movement of an electric charge from the rear frame to an earth via the placement surface is prevented. In addition, the rear frame is not connected to the earth, but is connected to a ground, that is, a wire electrically connected to each of a low-response detection circuit, a positive polarity high voltage power supply, and a negative polarity high voltage power supply.

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Patent No.	Application Date	Country	CPC
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) The present application claims foreign priority based on Japanese Patent Application No. 2022-142589, filed Sep. 7, 2022, the contents of which are incorporated herein by reference.
BACKGROUND OF THE INVENTION
1. Field of the Invention
(2) The invention relates to a technique for suppressing charging of a housing of a static eliminator that releases ions to an object for static elimination of the object.
2. Description of Related Art
(3) JP H10-289796 A discloses a static eliminator that applies positive and negative high voltages

to positive and negative electrode needles, respectively, to generate a corona discharge, thereby generating positive ions and negative ions. In order to reliably eliminate static electricity of an object by such a static eliminator, it is important to equally balance a generation amount of positive ions and a generation amount of negative ions. Therefore, this static eliminator includes a detection resistor that detects a current flowing between the static eliminator and an earth, and performs feedback control on the positive and negative high voltages applied to the positive and negative electrode needles based on a voltage generated in the detection resistor.

(4) As a result, it is possible to suppress a difference between the generation amount of positive ions and the generation amount of negative ions and to realize appropriate ion balance.

(5) Meanwhile, in order to suppress charging of a housing of the static eliminator, the housing can be partially or entirely formed with a conductive member. However, when the conductive member is short-circuited to the earth, the detection of the current flowing between the static eliminator and the earth is affected by movement of an electric charge from the conductive member to the earth, and there is a possibility that the ion balance cannot be appropriately controlled.

SUMMARY OF THE INVENTION

(6) The invention has been made in view of the above problem, and an object thereof is to provide a technique capable of suppressing charging of a housing of a static eliminator while avoiding an influence on control of ion balance.

(7) According to one embodiment of the invention, a static eliminator that releases ions to an object to eliminate static electricity of the object, the static eliminator including: an ion generator that generates a corona discharge in response to application of a positive polarity high voltage to generate positive ions, and generates a corona discharge in response to application of a negative polarity high voltage to generate negative ions; a high voltage application unit that applies the positive polarity high voltage and the negative polarity high voltage to the ion generator; a ground electrode short-circuited to an earth; a detection circuit that detects an ion current flowing between the earth and the static eliminator via the ground electrode; a feedback control unit that executes feedback control on the high voltage application unit to make the ion current detected by the detection circuit be a predetermined target value; a wire electrically connected to each of the detection circuit and the high voltage application unit; and a housing having a conductive member, which is insulated from a placement surface on which the static eliminator is placed and electrically connected to the wire, the housing accommodating the detection circuit.

(8) In the invention (static eliminator) configured as described above, the ion generator generating the positive ions and the negative ions, and the high voltage application unit applying the positive polarity high voltage and the negative polarity high voltage to the ion generator are provided. Then, the ion generator generates the positive ions when the high voltage application unit applies the positive polarity high voltage to the ion generator, and the ion generator generates the negative ions when the high voltage application unit applies the negative polarity high voltage to the ion generator. Further, the ion current flowing between the earth and the static eliminator via the ground electrode is detected, and the feedback control is executed on the high voltage application unit such that the ion current has the predetermined target value. The feedback control based on the ion current enables appropriate control of ion balance. Further, in order to suppress charging of the housing of the static eliminator, the housing is provided with the conductive member. The conductive member is insulated from the placement surface of the static eliminator, so that an electric charge can be prevented from moving from the conductive member to the earth via the placement surface. In addition, the conductive member is connected not to the earth but to a wire electrically connected to each of the detection circuit and the high voltage application unit. As a result, the electric charge of the conductive member is absorbed by the high voltage application unit, so that the electric charge can be prevented from moving from the conductive member to the earth. As a result, it is possible to suppress the charging of the housing of the static eliminator while avoiding an influence on the control of the ion balance.

(9) As described above, it is possible to suppress the charging of the housing of the static eliminator while avoiding the influence on the control of the ion balance according to the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a front perspective view illustrating an appearance of an example of a static eliminator according to the invention;
(2) FIG. **2** is a rear perspective view illustrating an appearance of the example of the static eliminator of FIG. **1**;
(3) FIG. **3** is an exploded perspective view of the example of the static eliminator of FIG. **1**;
(4) FIG. **4** is a rear view illustrating the inside of the static eliminator of FIG. **1**;
(5) FIG. **5**A is a rear view illustrating an example of a negative electrode unit;
(6) FIG. **5**B is a rear view illustrating an example of a positive electrode unit;
(7) FIG. **6**A is a rear perspective view illustrating a mode of fixing the negative electrode unit to a fixing base;
(8) FIG. **6**B is a rear perspective view illustrating a mode of fixing the positive electrode unit to the fixing base;
(9) FIG. **6**C is a rear perspective view illustrating a mode of fixing the negative electrode unit and the positive electrode unit to the fixing base;
(10) FIG. **6**D is an enlarged perspective view illustrating the mode of fixing the negative electrode unit and the positive electrode unit to the fixing base in an enlarged manner;
(11) FIG. **7**A is a perspective view illustrating a configuration in which a voltage is applied to the negative electrode unit;
(12) FIG. **7**B is a perspective view illustrating a configuration in which a voltage is applied to the positive electrode unit;
(13) FIG. **8**A is a rear view illustrating a configuration of a cleaning unit;
(14) FIG. **8**B is a perspective view illustrating the configuration of the cleaning unit;
(15) FIG. **9** is a lower perspective view illustrating a bottom surface of the static eliminator of FIG. **1**;
(16) FIG. **10** is a front perspective view illustrating the static eliminator in which a support fitting is attached to a housing;
(17) FIG. **11** is a front view illustrating the static eliminator in which the support fitting is attached to the housing;
(18) FIG. **12** is a cross-sectional view schematically illustrating a configuration of a fitting attachment portion for attaching the support fitting to the housing;
(19) FIG. **13** is a block diagram schematically illustrating a configuration of a controller which is an electrical equipment system of the static eliminator of FIG. **1**;
(20) FIG. **14** is a flowchart illustrating an example of an operation executed by the controller of FIG. **13**;
(21) FIG. **15**A is a block diagram illustrating details of an electrode unit controller;
(22) FIG. **15**B is a flowchart illustrating an example of voltage control executed in the operation of FIG. **14**;
(23) FIG. **16** is a perspective view schematically illustrating a modified example of the negative electrode unit and the positive electrode unit;
(24) FIG. **17** is a diagram schematically illustrating two systems that perform long-term feedback and short-term feedback; and
(25) FIG. **18** is a perspective view illustrating an example of an ion balance sensor.
DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(26) FIG. **1** is a front perspective view illustrating an appearance of an example of a static eliminator according to the invention; FIG. **2** is a rear perspective view illustrating an appearance of the example of the static eliminator of FIG. **1**; FIG. **3** is an exploded perspective view of the example of the static eliminator of FIG. **1**; and FIG. **4** is a rear view illustrating the inside of the static eliminator of FIG. **1**. Note that in the present specification, description will be given while appropriately indicating an X direction which is the horizontal direction, a Y direction which is the horizontal direction orthogonal to the X direction, and a Z direction which is the vertical direction. Further, one of both sides in the X direction is appropriately referred to as a front side Xf, and the other side is appropriately referred to as a rear side Xb.
(27) A static eliminator **1** includes a front cover **11**, a housing **2**, a fan unit **3**, a fixing base **4**, a negative electrode unit **5**, a positive electrode unit **6**, a cleaning unit **7**, and a rear cover **12**. The housing **2** is roughly divided into an upper part **2**U and a lower part **2**L provided on the lower side of the upper part **2**U. An accommodation chamber **201** is provided in the upper part **2**U of the housing **2**, and an electrical equipment accommodating portion **202** is provided in the lower part **2**L of the housing **2**. The accommodation chamber **201** has a rectangular shape as viewed from the X direction and is open in the X direction. The fan unit **3**, the fixing base **4**, the negative electrode unit **5**, the positive electrode unit **6**, and the cleaning unit **7** are arrayed in the X direction and accommodated in the accommodation chamber **201**. The electrical equipment accommodating portion **202** accommodates an electrical equipment system of the static eliminator **1**. Further, the front cover **11** is attached to the housing **2** from the front side Xf so as to oppose the accommodation chamber **201**, and the rear cover **12** is attached to the housing **2** from the rear side Xb so as to oppose the accommodation chamber **201**.
(28) The housing **2** includes a front frame **21** and a rear frame **25** provided on the rear side Xb of the front frame **21**. The front frame **21** and the rear frame **25** are arrayed in the X direction and attached to each other. The front frame **21** and the rear frame **25** are made of an antistatic resin and are electrically conductive. The antistatic resin can be formed by kneading an antistatic agent into a resin or coating a surface of a resin with an antistatic agent. The antistatic resin in the present embodiment is a resin having such a resistance value that an electric charge generated on a surface of the housing **2** flows to a ground G in a relatively short time, for example, several seconds when the housing **2** is made of the resin. An experimental result in which the electric charge generated on the surface of the housing **2** flows to the ground G in several seconds has been obtained when the housing **2** is made of a resin having a resistance value in the range of 10.sup.9Ω to 10.sup.12Ω. Further, it is sufficient that most of an outer surface of the housing **2** is made of the antistatic resin. In the present embodiment, a display section **23** is not made of the antistatic resin, but charging of a part of the housing **2** has a small influence.
(29) The front frame **21** includes a main frame **22** and the display section **23** provided on the front side Xf of the main frame **22**. The main frames **22** and the display section **23** are arrayed in the X direction and attached to each other. The main frame **22** is open in the X direction. The display section **23** is provided in an opening of the main frame **22** in the lower part **2**L and is arranged so as to be visually recognizable from the front side Xf. That is, the opening of the main frame **22** in a range of the upper part **2**U constitutes a part of the accommodation chamber **201**. Further, the main frame **22** in a range of the lower part **2**L constitutes a part of the electrical equipment accommodating portion **202**.
(30) The rear frame **25** is open in the X direction. An opening of the rear frame **25** in a range of upper part **2**U constitutes a part of the accommodation chamber **201**. Further, the rear frame **25** in a range of the lower part **2**L constitutes a part of the electrical equipment accommodating portion **202**.
(31) The front cover **11** includes a cover frame **111** made of an antistatic resin, and the cover frame **111** is attached to the front frame **21** of the housing **2** from the front side Xf in the upper part **2**U. The cover frame **111** covers the accommodation chamber **201** from the front side Xf. Further, the

cover frame **111** includes a mesh portion **112** provided with a plurality of slits, and the mesh portion **112** opposes the accommodation chamber **201** from the front side Xf. Further, a front wire mesh **115** (metal mesh) having a circular shape as viewed from the X direction is attached to the front frame **21**. The front wire mesh **115** opposes the accommodation chamber **201** from the front side Xf and opposes the mesh portion **112** from the rear side Xb. The mesh portion **112** and the front wire mesh **115** allow passage of air in the X direction. Note that the cover frame **111** has the mesh portion **112** provided with the plurality of slits in the present embodiment, but may have any shape that can guide air generated by a fan **33**, which will be described later, to a desired region. Further, the front cover **11** is attached to the housing **2**, but a configuration in which the front cover **11** selected from a plurality of the front covers **11** having different shapes of the cover frame **111** is attached to the housing **2** may be adopted. According to this configuration, a user can attach the front cover **11** selected according to a use environment of the static eliminator **1** to the housing **2**. For example, it is possible to attach the front cover **11** suitable for guiding the air to the vicinity in a case where a distance between the static eliminator **1** and an object to be neutralized is short and to attach the front cover **11** suitable for guiding the air far away in a case where the distance between the static eliminator **1** and the object to be neutralized is long. Furthermore, in the configuration in which the front cover **11** is switchable, a parameter regarding the operation of the static eliminator **1** may be set according to a type of the front cover **11** attached to the housing **2**.

(32) The rear cover **12** includes a cover frame **121** made of an antistatic resin, and the cover frame **121** is attached to the rear frame **25** of the housing **2** from the rear side Xb in the upper part **2**U. The cover frame **111** has an opening **122** having a circular shape as viewed from the X direction, and the opening **122** opposes the accommodation chamber **201** from the rear side Xb. Furthermore, the rear cover **12** includes a rear wire mesh **125** (metal mesh) having a circular shape as viewed from the X direction. The rear wire mesh **125** is fitted into the opening **122** and attached to the cover frame **121**, and opposes the accommodation chamber **201** from the rear side Xb. The rear wire mesh **125** allows passage of air in the X direction. Further, the rear wire mesh **125** is short-circuited to the ground G (FIG. **9**). Note that a mode of electrically connecting the rear wire mesh **125** and the ground G is not limited to the short circuit, and these may be connected via a resistor.

(33) The fan unit **3** is arranged in the accommodation chamber **201** of the housing **2** and is located on the rear side Xb of the front wire mesh **115** of the front cover **11**. The fan unit **3** includes a support frame **31** having a rectangular shape as viewed from the X direction, and the support frame **31** is arranged in the accommodation chamber **201** and attached to the housing **2**. In the support frame **31**, a ventilation opening **32** having a circular shape as viewed from the X direction is open in the X direction. The ventilation opening **32** opposes the front wire mesh **115** of the front cover **11** from the rear side Xb. Furthermore, the fan unit **3** includes the fan **33** having a circular shape as viewed from the X direction. The fan **33** includes a rotating shaft **331** provided parallel to the X direction and a plurality of blades **332** provided around the rotating shaft **331**. Further, the fan **33** is arranged in the ventilation opening **32** of the support frame **31** and opposes the front wire mesh **115** of the front cover **11** from the rear side Xb. The fan **33** is supported by the support frame **31** so as to be rotatable about a rotation center parallel to the X direction, and rotates about the rotation center, thereby generating air (in other words, air flow) in an air blowing direction Dw from the rear side Xb toward the front side Xf in the X direction.

(34) The fixing base **4** is arranged in the accommodation chamber **201** of the housing **2** and is located on the rear side Xb of the fan unit **3**. The fixing base **4** includes a fixing frame **41** having a rectangular shape as viewed from the X direction, and the fixing frame **41** is arranged in the accommodation chamber **201** and attached to the housing **2**. In the fixing frame **41**, a ventilation opening **42** is open in the X direction. The ventilation opening **42** has a rectangular shape whose four corners are cut out in an arc shape as viewed from the X direction. Further, the fixing base **4** includes fixing portions **43**, **44**, **45**, and **46** provided at four corners of the fixing frame **41**. The fixing portions **43**, **44**, **45**, and **46** are located on the outer side of the four corners of the ventilation

opening **42**, respectively. Furthermore, the fixing base **4** has an I-shaped part that supports the cleaning unit **7** with respect to the fixing frame **41** as will be described later.

(35) The negative electrode unit **5** is arranged in the accommodation chamber **201** of the housing **2** and is fixed to the fixing frame **41** of the fixing base **4** from the rear side Xb. The negative electrode unit **5** has a configuration illustrated in FIG. **5**A. FIG. **5**A is a rear view illustrating an example of the negative electrode unit. FIG. **5**A illustrates a virtual circle Cv (a circle indicated by a broken line) having a circular shape centered on a center point Pc as viewed from the X direction, and a circumferential direction Dc centered on the center point Pc.

(36) As illustrated in FIG. **5**A, the negative electrode unit **5** includes a first unit frame **51** provided along the virtual circle Cv. In other words, the first unit frame **51** has an arc shape along the virtual circle Cv. Furthermore, the negative electrode unit **5** has a plurality of (four) electrode needles Nm arrayed at a constant array pitch (90 degrees) in the circumferential direction Dc along the virtual circle Cv. The plurality of electrode needles Nm are arrayed along an inner wall **511** of the first unit frame **51** and protrude inward (in other words, to the center point Pc side of the virtual circle Cv) from the inner wall **511**. A cable (wire) electrically connected to each of the electrode needles Nm is built in the first unit frame **51**, and a voltage is applied to each of the electrode needles Nm through the cable.

(37) Further, the negative electrode unit **5** has a plurality of (four) fixing portions **53**, **54**, **55**, and **56** arrayed at a constant array pitch (90 degrees) in the circumferential direction Dc. In this example, the number of the electrode needles Nm is equal to the number of the fixing portions **53**, **54**, **55**, and **56**. The plurality of fixing portions **53**, **54**, **55**, and **56** are arrayed along an outer wall **512** of the first unit frame **51**, and protrude outward (in other words, to the opposite side of the center point Pc of the virtual circle Cv) from the outer wall **512**. In the circumferential direction Dc, a phase of the array of the plurality of fixing portions **53**, **54**, **55**, and **56** is shifted from a phase of the array of the plurality of electrode needles Nm. That is, the fixing portions **53**, **54**, **55**, and **56** are provided at positions shifted from the electrode needles Nm in the circumferential direction Dc. The fixing portions **53**, **54**, **55**, and **56** are respectively fastened to the fixing portions **43**, **44**, **45**, and **46** of the fixing base **4** by screws S.

(38) The air generated by the fan **33** of the fan unit **3** described above passes through a flow path Fw surrounded by the first unit frame **51** of the negative electrode unit **5** in the air blowing direction Dw. In other words, the first unit frame **51** of the negative electrode unit **5** has a curved shape (arc shape) so as to surround the flow path Fw through which the air generated by the fan **33** passes.

(39) As illustrated in FIG. **3**, the positive electrode unit **6** is arranged in the accommodation chamber **201** of the housing **2** and is fixed to the fixing frame **41** of the fixing base **4** from the rear side Xb. The positive electrode unit **6** has a configuration illustrated in FIG. **5**B. FIG. **5**B is a rear view illustrating an example of the positive electrode unit. FIG. **5**B illustrates the virtual circle Cv and the circumferential direction Dc similarly to FIG. **5**A.

(40) As illustrated in FIG. **5**B, the positive electrode unit **6** includes a second unit frame **61** provided along the virtual circle Cv. In other words, the second unit frame **61** has an arc shape along the virtual circle Cv. Furthermore, the positive electrode unit **6** has a plurality of (four) electrode needles Np arrayed at a constant array pitch (90 degrees) in the circumferential direction Dc along the virtual circle Cv. The plurality of electrode needles Np are arrayed along an inner wall **611** of the second unit frame **61** and protrude inward (in other words, to the center point Pc side of the virtual circle Cv) from the inner wall **611**. A cable (wire) electrically connected to each of the electrode needles Np is built in the second unit frame **61**, and a voltage is applied to each of the electrode needles Np through the cable.

(41) Further, the positive electrode unit **6** has a plurality of (four) fixing portions **63**, **64**, **65**, and **66** arrayed at a constant array pitch (90 degrees) in the circumferential direction Dc. In this example, the number of the electrode needles Np is equal to the number of the fixing portions **63**, **64**, **65**, and

66. The plurality of fixing portions **63**, **64**, **65**, and **66** are arrayed along an outer wall **612** of the second unit frame **61**, and protrude outward (in other words, to the opposite side of the center point Pc of the virtual circle Cv) from the outer wall **612**. In the circumferential direction Dc, a phase of the array of the plurality of fixing portions **63**, **64**, **65**, and **66** is shifted from a phase of the array of the plurality of electrode needles Np. That is, the fixing portions **63**, **64**, **65**, and **66** are provided at positions shifted from the electrode needles Np in the circumferential direction Dc. The fixing portions **63**, **64**, **65**, and **66** are respectively fastened to the fixing portions **43**, **44**, **45**, and **46** of the fixing base **4** by screws S.

(42) The air generated by the fan **33** of the fan unit **3** described above passes through the flow path Fw surrounded by the second unit frame **61** of the positive electrode unit **6** in the air blowing direction Dw. In other words, the second unit frame **61** of the positive electrode unit **6** has a curved shape (arc shape) so as to surround the flow path Fw through which the air generated by the fan **33** passes.

(43) The negative electrode unit **5** and the positive electrode unit **6** are arrayed in the X direction in the accommodation chamber **201**, and the positive electrode unit **6** is arranged on the rear side Xb of the negative electrode unit **5**. Further, the negative electrode unit **5** and the positive electrode unit **6** are fixed to the fixing base **4** such that the first unit frame **51** of the negative electrode unit **5** and the second unit frame **61** of the positive electrode unit **6** overlap each other as viewed from the X direction. It is sufficient that the fixing base **4** is a member that fixes the negative electrode unit **5** and the positive electrode unit **6** so as to have a desired arrangement relationship, and the fixing base **4** may be configured using a single member or a plurality of members. Further, another member such as a member constituting the housing **2** may also be configured to serve as the fixing base **4**.

(44) FIG. **6**A is a rear perspective view illustrating a mode of fixing the negative electrode unit to the fixing base; FIG. **6**B is a rear perspective view illustrating a mode of fixing the positive electrode unit to the fixing base; FIG. **6**C is a rear perspective view illustrating a mode of fixing the negative electrode unit and the positive electrode unit to the fixing base; and FIG. **6**D is an enlarged perspective view illustrating the mode of fixing the negative electrode unit and the positive electrode unit to the fixing base in an enlarged manner.

(45) The fixing portion **43** has a protruding plate **431** protruding outward from the first and second unit frames **51** and **61** as viewed from the X direction. The protruding plate **431** protrudes to the upper left side from the first and second unit frames **51** and **61** in a rear view. Furthermore, the fixing portion **43** includes a fastening portion **432** protruding from the protruding plate **431** to the rear side Xb in the X direction and a fastening portion **433** protruding from the protruding plate **431** to the rear side Xb in the X direction. In the fastening portion **432**, a screw hole **432***h* extending in the X direction is open to the rear side Xb. In the fastening portion **433**, a screw hole **433***h* extending in the X direction is open to the rear side Xb. The screws S are screwed into the screw holes **432***h* and **433***h*. In the circumferential direction Dc, the fastening portion **432** and the fastening portion **433** are provided to be shifted from each other, and the fastening portion **432** is located on one side (clockwise side in the rear view) of the fastening portion **433**.

(46) The fixing portion **44** has a protruding plate **441** protruding outward from the first and second unit frames **51** and **61** as viewed from the X direction. The protruding plate **441** protrudes to the lower left side from the first and second unit frames **51** and **61** in the rear view. Furthermore, the fixing portion **44** includes a fastening portion **442** protruding from the protruding plate **441** to the rear side Xb in the X direction and a fastening portion **443** protruding from the protruding plate **441** to the rear side Xb in the X direction. In the fastening portion **442**, a screw hole **442***h* extending in the X direction is open to the rear side Xb. In the fastening portion **443**, a screw hole **443***h* extending in the X direction is open to the rear side Xb. The screws S are screwed into the screw holes **442***h* and **443***h*. In the circumferential direction Dc, the fastening portion **442** and the fastening portion **443** are provided to be shifted from each other, and the fastening portion **442** is

located on one side (clockwise side in the rear view) of the fastening portion **443**.
(47) The fixing portion **45** has a protruding plate **451** protruding outward from the first and second unit frames **51** and **61** as viewed from the X direction. The protruding plate **451** protrudes to the lower right side from the first and second unit frames **51** and **61** in the rear view. Furthermore, the fixing portion **45** includes a fastening portion **452** protruding from the protruding plate **451** to the rear side Xb in the X direction and a fastening portion **453** protruding from the protruding plate **441** to the rear side Xb in the X direction. In the fastening portion **452**, a screw hole **452***h* extending in the X direction is open to the rear side Xb. In the fastening portion **453**, a screw hole **453***h* extending in the X direction is open to the rear side Xb. The screws S are screwed into the screw holes **452***h* and **453***h*. In the circumferential direction Dc, the fastening portion **452** and the fastening portion **453** are provided to be shifted from each other, and the fastening portion **452** is located on one side (clockwise side in the rear view) of the fastening portion **453**.
(48) The fixing portion **46** has a protruding plate **461** protruding outward from the first and second unit frames **51** and **61** as viewed from the X direction. The protruding plate **461** protrudes to the upper right side from the first and second unit frames **51** and **61** in the rear view. Furthermore, the fixing portion **46** includes a fastening portion **462** protruding from the protruding plate **461** to the rear side Xb in the X direction and a fastening portion **463** protruding from the protruding plate **441** to the rear side Xb in the X direction. In the fastening portion **462**, a screw hole **462***h* extending in the X direction is open to the rear side Xb. In the fastening portion **463**, a screw hole **463***h* extending in the X direction is open to the rear side Xb. The screws S are screwed into the screw holes **462***h* and **463***h*. In the circumferential direction Dc, the fastening portion **462** and the fastening portion **463** are provided to be shifted from each other, and the fastening portion **462** is located on one side (clockwise side in the rear view) of the fastening portion **463**.
(49) The fixing portions **53**, **54**, **55**, and **56** of the negative electrode unit **5** are respectively fastened to the fastening portions **432**, **442**, **452**, and **462** of the fixing base **4** with the screws S, respectively. Specifically, an insertion hole extending in the X direction is opened in the fixing portion **53**. Then, the screw S inserted into the insertion hole of the fixing portion **53** is screwed into the screw hole **432***h* of the fastening portion **432** in a state in which the insertion hole of the fixing portion **53** adjacent to the fastening portion **432** from the rear side Xb opposes the screw hole **432***h* of the fastening portion **432** in the X direction. Thus, the fixing portion **53** is fastened to the fastening portion **432**. Further, the fixing portions **54**, **55**, and **56** are similarly fastened.
(50) The fixing portions **63**, **64**, **65**, and **66** of the positive electrode unit **6** are fastened to the fastening portions **433**, **443**, **453**, and **463** of the fixing base **4** with the screws S, respectively. Specifically, an insertion hole extending in the X direction is opened in the fixing portion **63**. Then, the screw S inserted into the insertion hole of the fixing portion **63** is screwed into the screw hole **433***h* of the fastening portion **433** in a state in which the insertion hole of the fixing portion **63** adjacent to the fastening portion **433** from the rear side Xb opposes the screw hole **433***h* of the fastening portion **433** in the X direction. Thus, the fixing portion **63** is fastened to the fastening portion **433**. Further, the fixing portions **64**, **65**, and **66** are similarly fastened.
(51) Incidentally, the fastening portions **433**, **443**, **453**, and **463** have the same length, and the fastening portions **432**, **442**, **452**, and **462** have the same length. On the other hand, the fastening portions **433**, **443**, **453**, and **463** are longer than the fastening portions **432**, **442**, **452**, and **462**. Therefore, the positive electrode unit **6** fastened to the fastening portions **433**, **443**, **453**, and **463** is located on the rear side Xb of the negative electrode unit **5** fastened to the fastening portions **432**, **442**, **452**, and **462**. In particular, the lengths of the fastening portions **433**, **443**, **453**, and **463** and the fastening portions **432**, **442**, **452**, and **462** are set such that a gap is formed between the negative electrode unit **5** and the positive electrode unit **6** in the X direction.
(52) Further, the number of the electrode needles Nm included in the negative electrode unit **5** and the number of the electrode needles Np included in the positive electrode unit **6** are equal (four), and the array pitch of the electrode needles Nm in the negative electrode unit **5** and the array pitch

of the electrode needles Np in the positive electrode unit **6** are equal (90 degrees). On the other hand, for example, as illustrated in FIG. **4**, a phase of the array of the plurality of electrode needles Nm in the negative electrode unit **5** and a phase of the array of the plurality of electrode needles Np in the positive electrode unit **6** are shifted by 45 degrees. Therefore, the electrode needles Np and the electrode needles Nm are alternately arrayed at a half pitch (45 degrees) that is half the array pitch as viewed from the X direction. The electrode needles Np and the electrode needles Nm are arrayed in the circumferential direction Dc so as to surround the flow path Fw of the air flowing in the air blowing direction Dw generated by the fan **33**, and tip portions of the electrode needles Np and the electrode needles Nm protrude to the flow path Fw.
(53) FIG. **7**A is a perspective view illustrating a configuration in which a voltage is applied to the negative electrode unit. The static eliminator **1** has a harness Hm, which extends from the electrical equipment system accommodated in the electrical equipment accommodating portion **202** to the fixing portion **55** of the negative electrode unit **5**, and an electrode terminal is exposed at a tip of the harness Hm. Further, an electrode terminal of the cable electrically connected to the electrode needles Nm is exposed on a side surface on the front side Xf of the fixing portion **55**. Then, the fixing portion **55** is fastened to the fastening portion **452** in a state in which the electrode terminal of the harness Hm is sandwiched between the fastening portion **452** and the electrode terminal of the fixing portion **55** of the negative electrode unit **5**. As a result, the electrode terminal of the harness Hm and the electrode terminal of the cable of the negative electrode unit **5** are electrically in contact with each other, and a voltage supplied from the electrical equipment system via the harness Hm is applied to the electrode needles Nm of the negative electrode unit **5**.
(54) FIG. **7**B is a perspective view illustrating a configuration in which a voltage is applied to the positive electrode unit. The static eliminator **1** has a harness Hp, which extends from the electrical equipment system accommodated in the electrical equipment accommodating portion **202** to the fixing portion **64** of the positive electrode unit **6**, and an electrode terminal is exposed at a tip of the harness Hp. Further, an electrode terminal of the cable electrically connected to the electrode needles Np is exposed on a side surface on the front side Xf of the fixing portion **64**. Then, the fixing portion **64** is fastened to the fastening portion **443** in a state in which the electrode terminal of the harness Hp is sandwiched between the fastening portion **443** and the electrode terminal of the fixing portion **64** of the positive electrode unit **6**. As a result, the electrode terminal of the harness Hp and the electrode terminal of the cable of the positive electrode unit **6** are electrically in contact with each other, and a voltage supplied from the electrical equipment system via the harness Hp is applied to the electrode needles Np of the positive electrode unit **6**.
(55) FIG. **8**A is a rear view illustrating a configuration of the cleaning unit, and FIG. **8**B is a perspective view illustrating the configuration of the cleaning unit. The cleaning unit **7** includes cleaning brushes **71***m* and **71***p*, a motor **72**, a rotating plate **73** driven by the motor **72**, and a brush supporter **74** that supports the cleaning brushes **71***m* and **71***p* with respect to the rotating plate **73**.
(56) The motor **72** is accommodated in a cylindrical part of the fixing base **4** centered on an axis parallel to the X direction. The rotating plate **73** has a disk shape centered on the axis. Further, the motor **72** and the rotating plate **73** are arranged at the center of the virtual circle Cv as viewed from the X direction, and a clearance CL is provided between each of the inner walls **511** and **611** of the first and second unit frames **51** and **61** and each of outer circumferences of the motor **72** and the rotating plate **73**. This clearance CL opposes the plurality of blades **332** of the fan **33**, and the air generated by the fan **33** passes through the clearance CL in the flow path Fw. The motor **72** has a rotating shaft passing through the center point Pc and parallel to the X direction, and the rotating plate **73** is provided coaxially with the motor **72**. The rotating plate **73** is driven by the motor **72** to rotate in the circumferential direction Dc about the rotating shaft of the motor **72**. In this example, the motor **72** is a stepping motor. However, a type of the motor **72** is not limited to this example.
(57) The brush supporter **74** includes an attachment portion **741** attached to a back surface of the rotating plate **73**, and a screw **742** for fastening the attachment portion **741** to the back surface of

the rotating plate **73**. A tip of the attachment portion **741** protrudes to the outer side of the rotating plate **73**, and the brush supporter **74** includes an extending portion **743** extending from a tip of the rotating plate **73** to the front side Xf in the X direction, and two support portions **744***m* and **744***p* protruding from the extending portion **743** to the outer side in the radial direction around the center point Pc. Each of the support portions **744***m* and **744***p* extends from the extending portion **743** to the outer side of the rotating plate **73** in the radial direction. The support portions **744***m* and **744***p* are arrayed in the X direction, and the support portion **744***p* is located on the rear side Xb of the support portion **744***m*. Furthermore, the brush supporter **74** includes the brush holders **745***m*, **745***p* attached tips of the support portions **744***m* and **744***p*, respectively. The brush holders **745***m* and **745***p* are arrayed in the X direction, and the brush holder **745***p* is located on the rear side Xb of the brush holder **745***m*.

(58) The cleaning brush **71***m* is held by the brush holder **745***m*, and the cleaning brush **71***p* is held by the brush holder **745***p*. The cleaning brush **71***m* and the cleaning brush **71***p* are provided to correspond to the electrode needles Nm and the electrode needles Np, respectively, and extend in the radial direction around the center point Pc. The cleaning brush **71***m* and the cleaning brush **71***p* are arrayed in the X direction, and the cleaning brush **71***p* is located on the rear side Xb of the cleaning brush **71***m*. The cleaning brush **71***m* opposes the inner wall **511** of the first unit frame **51**, and the cleaning brush **71***p* opposes the inner wall **611** of the second unit frame **61**. In such a configuration, the cleaning brushes **71***m* and **71***p* move in the circumferential direction Dc by a driving force of the motor **72**. Then, the cleaning unit **7** cleans the electrode needles Nm and Np as follows by driving the cleaning brushes **71***m* and **71***p* by the motor **72**.

(59) That is, a plurality of cleaning positions Lm arrayed in the circumferential direction Dc are provided, and the plurality of cleaning positions Lm correspond to the plurality of electrode needles Nm, respectively. Then, the cleaning brush **71***m* is located at one cleaning position Lm corresponding to one electrode needle Nm to be cleaned among the plurality of electrode needles Nm, thereby coming into contact with the one electrode needle Nm. In particular, the motor **72** causes the cleaning brush **71***m* in contact with one electrode needle Nm at one cleaning position Lm to slightly reciprocate in the circumferential direction Dc, whereby dirt adhering to the one electrode needle Nm can be scraped off by a tip of the cleaning brush **71***m*.

(60) Similarly, a plurality of cleaning positions Lp arrayed in the circumferential direction Dc are provided, and the plurality of cleaning positions Lp correspond to the plurality of electrode needles Np, respectively. Then, the cleaning brush **71***p* is located at one cleaning position Lp corresponding to one electrode needle Np to be cleaned among the plurality of electrode needles Np, thereby coming into contact with the one electrode needle Np. In particular, the motor **72** causes the cleaning brush **71***p* in contact with one electrode needle Np at one cleaning position Lp to slightly reciprocate in the circumferential direction Dc, whereby dirt adhering to the one electrode needle Np can be scraped off by a tip of the cleaning brush **71***p*.

(61) Further, the cleaning unit **7** also includes a brush cleaner **75** that cleans the cleaning brushes **71***m* and **71***p*. The brush cleaner **75** includes an accommodation box **751** that accommodates the cleaning brushes **71***m* and **71***p*. The accommodation box **751** is open in the circumferential direction Dc (in other words, the Y direction), and the cleaning brushes **71***m* and **71***p* can be put into the accommodation box **751** or taken out from the accommodation box **751** by moving the cleaning brushes **71***m* and **71***p* in the circumferential direction Dc by the motor **72**. FIGS. **8**A and **8**B illustrate a state in which the cleaning brushes **71***m* and **71***p* are taken out of the accommodation box **751**, and FIG. **4** illustrates a state in which the cleaning brushes **71***m* and **71***p* are put into the accommodation box **751**.

(62) The brush cleaner **75** removes dirt from the cleaning brushes **71***m* and **71***p* by sliding contact members provided in the accommodation box **751**. That is, in the accommodation box **751**, the sliding contact members are provided, respectively, to correspond to openings on both sides in the circumferential direction Dc of the accommodation box **751**. Then, the tips of the cleaning brushes

71*m* and **71***p* moving in the circumferential direction Dc by the driving force of the motor **72** are slid on the sliding contact members of the brush cleaner **75**. As a result, the dirt adhering to the cleaning brushes **71***m* and **71***p* is scraped off against by the sliding contact members of the brush cleaner **75**, whereby cleaning of the cleaning brushes **71***m* and **71***p* is executed. This cleaning is executed when the cleaning brushes **71***m* and **71***p* enter the accommodation box **751** and exit the accommodation box **751**.

(63) The cleaning unit **7** is supported by the I-shaped part of the fixing base **4** described above. Specifically, the motor **72** is supported by the fixing base **4** at the center of the I-shaped part. Further, the brush cleaner **75** is attached to a part having a flat plate-shape in a bottom portion of the fixing base **4**.

(64) Next, a mechanism for supporting the housing **2** with respect to a placement surface on which the static eliminator **1** is placed will be described. FIG. **9** is a lower perspective view illustrating a bottom surface of the static eliminator of FIG. **1**. The static eliminator **1** includes insulating pads **131**, **132**, **133**, and **134** provided at four corners of a bottom surface **2**B of the housing **2**. Among them, the insulating pad **131** and the insulating pad **132** are arranged on the bottom surface of a cover plate **23** of the front frame **21** with a clearance in the Y direction. Further, the insulating pad **133** and the insulating pad **134** are arranged on a bottom surface of an attachment frame **27** of the rear frame **25** with a clearance in the Y direction. The insulating pads **131**, **132**, **133**, and **134** protrude downward from the bottom surface **2**B of the housing **2**. Therefore, the insulating pads **131**, **132**, **133**, and **134** are in contact with the placement surface between the bottom surface **2**B of the housing **2** and the placement surface in a state in which the housing **2** is placed on the placement surface, thereby separating the front frame **21** and the rear frame **25** from the placement surface.

(65) Further, in addition to the insulating pads **131**, **132**, **133**, and **132**, the static eliminator **1** includes a support fitting that supports the housing **2** with respect to the placement surface (FIGS. **10**, **11**, and **12**). FIG. **10** is a front perspective view illustrating the static eliminator in which the support fitting is attached to the housing; FIG. **11** is a front view illustrating the static eliminator in which the support fitting is attached to the housing; and FIG. **12** is a cross-sectional view schematically illustrating a configuration of a fitting attachment portion for attaching the support fitting to the housing. Note that FIG. **11** illustrates a placement surface Axy which is a horizontal plane, and FIG. **12** illustrates an axis Ay which is a virtual straight line parallel to the Y direction.

(66) The static eliminator **1** illustrated in FIGS. **10** and **11** includes a support fitting **14** that is made of metal and is electrically conductive, and two fitting attachment portions **15** for detachably attaching the support fitting **14** to the housing **2**. The support fitting **14** includes a placement plate **141** extending parallel to the Y direction and mounted on the placement surface Axy, and two upright plates **142** extending upward from both ends of the placement plate **141** in the Y direction. In the Y direction, the housing **2** is located between the two upright plates **142**, and each of the upright plates **142** opposes a side surface of the housing **2** in the Y direction with a clearance therebetween.

(67) The two fitting attachment portions **15** are provided to correspond to the two upright plates **142**, respectively, and each of the fitting attachment portions **15** attaches the corresponding upright plate **142** to the side surface of the housing **2** in the Y direction. That is, an upper end portion **143** of the upright plate **142** opposes the rear frame **25** in the Y direction, and is attached to the rear frame **25** by the fitting attachment portion **15**. As described above, the rear frame **25** is a part of the rear frame **25** made of an antistatic resin, and is electrically conductive.

(68) Further, details of a configuration in which the upright plate **142** is attached to the rear frame **25** by the fitting attachment portion **15** are as illustrated in FIG. **12**. Note that, in FIG. **12**, a member (the rear frame **25**) with a light dot pattern is an antistatic resin, members (an inner spacer **16**, an outer spacer **17**, and a resin sheet **192**) with a dark dot pattern are insulators, and members (a screw **18**, washers **191** and **193**, and a nut **194**) with oblique hatching are metal.

(69) Aside surface of the rear frame **25** includes a flat plate portion **261** parallel to the Z direction and orthogonal to the Y direction, and a protruding portion **262** protruding outward from the flat plate portion **261** in the Y direction. An outer shape of the protruding portion **262** is a truncated cone centered on the axis Ay having a diameter decreasing outward. Further, the rear frame **25** is provided with a through hole **263** penetrating through the flat plate portion **261** and the protruding portion **262** in the Y direction. The through holes **263** includes spaces **263***a*, **263***b*, **263***c*, and **263***d* each of which has the axis Ay as the center. The spaces **263***a*, **263***b*, **263***c*, and **263***d* are aligned in the Y direction in this order from the outer side to the inner side of the housing **2**. That is, in the Y direction, the space **263***b* is provided on the inner side of the space **263***a*, the space **263***c* is provided on the inner side of the space **263***b*, and the space **263***d* is provided on the inner side of the space **263***c*. Further, a diameter of the space **263***b* is smaller than a diameter of the space **263***a*, a diameter of the space **263***c* is larger than the diameter of the space **263***b*, and a diameter of the space **263***d* is larger than the diameter of the space **263***c*.
(70) The upper end portion **143** of the upright plate **142** opposes the protruding portion **262** from the outer side in the Y direction. The upper end portion **143** is provided with a through hole **144** penetrating therethrough in the Y direction. The through hole **144** includes spaces **144***a* and **144***b* each of which has the axis Ay as the center. The spaces **144***a* and **144***b* are aligned in the Y direction in this order from the outer side to the inner side of the housing **2**. That is, the space **144***b* is provided on the inner side of the space **144***a* in the Y direction. Further, a diameter of the space **144***b* is larger than a diameter of the space **144***a*.
(71) On the other hand, the fitting attachment portion **15** has an inner spacer **16** that has an insulating property and is arranged between the upper end portion **143** of the upright plate **142** and the rear frame **25** in the Y direction. An outer shape of the inner spacer **16** is a cylindrical shape having the same diameter as the space **144***b* of the through hole **144**. Further, the inner spacer **16** has a through hole **161** penetrating therethrough in the Y direction. The through hole **161** includes spaces **161***a*, **161***b*, and **161***c* each of which has the axis Ay as the center. The spaces **161***a*, **161***b*, and **161***c* are aligned in the Y direction in this order from the outer side to the inner side of the housing **2**. That is, in the Y direction, the space **161***b* is provided on the inner side of the space **161***a*, and the space **161***c* is provided on the inner side of the space **161***b*. Further, a diameter of the space **161***b* is larger than a diameter of the space **161***a*, and diameters of both end surfaces of the space **161***c* are larger than the diameter of the space **161***b*. Note that the diameter of the space **161***c* decreases outward.
(72) The protruding portion **262** of the rear frame **25** is fitted into the space **161***c* of the through hole **161** of the inner spacer **16**. Furthermore, the inner spacer **16** is fitted into the space **144***b* of the through hole **144** of the upright plate **142**. As a result, the rear frame **25**, the inner spacer **16**, and the upright plate **142** are positioned to each other, and the through hole **263** of the rear frame **25**, the through hole **161** of the inner spacer **16**, and the through hole **144** of the upright plate **142** oppose each other in the Y direction.
(73) Furthermore, the fitting attachment portion **15** further includes an outer spacer **17** that has an insulating property and is provided on the outer side of the upright plate **142** in the Y direction. The outer spacer **17** includes a spacer body **171** and a protruding portion **172** protruding inward from the spacer body **171** in the Y direction. An outer shape of the protruding portion **172** is a cylindrical shape centered on the axis Ay. Further, the outer spacer **17** is provided with a through hole **173** penetrating through the spacer body **171** and the protruding portion **172** in the Y direction. The through hole **173** includes spaces **173***a* and **173***b* each of which has the axis Ay as the center. The spaces **173***a* and **173***b* are aligned in the Y direction in this order from the outer side to the inner side of the housing **2**. That is, the space **173***b* is provided on the inner side of the space **173***a* in the Y direction. Further, a diameter of the space **173***b* is smaller than a diameter of the space **173***a*.
(74) The protruding portion **172** of the outer spacer **17** is fitted into the space **144***a* of the through hole **144** of the upright plate **142** and the space **161***a* of the through hole **161** of the inner spacer **16**.

As a result, the outer spacer **17** is positioned with respect to the upright plate **142** and the inner spacer **16**, and the through hole **161** of the inner spacer **16** and the through hole **173** of the outer spacer **17** oppose each other in the Y direction. Note that the protruding portion **172** is loosely fitted to the through hole **144** and the through hole **161**.

(75) In this manner, the through hole **263** of the rear frame **25**, the through hole **161** of the inner spacer **16**, the through hole **144** of the upright plate **142**, and the through hole **173** of the outer spacer **17** are aligned in the Y direction about the axis Ay as the center. On the other hand, the fitting attachment portion **15** has the screw **18** that is made of metal and is inserted into the through holes **263** and **161** and the through holes **144** and **173** from the outer side. The screw **18** includes a shaft portion **181** provided with a screw groove and a head portion **182** provided at one end of the shaft portion **181**. The shaft portion **181** is inserted into the through holes **263**, **161**, **144**, and **173** in a state in which the shaft portion **181** is parallel to the Y direction and the head portion **182** faces outward. On the other hand, the fitting attachment portion **15** has the three washers **191**, **192**, and **193** and the nut **194** each of which is made of metal. The washer **191** is arranged in the space **173***a* of the through hole **173** of the outer spacer **17**, the washer **192** is arranged in the space **161***b* of the through hole **161** of the inner spacer **16**, the washer **193** is arranged in the space **263***a* of the through hole **263** of the rear frame **25**, and the nut **194** is arranged in the space **263***c* of the through hole **263** of the rear frame **25**. Then, the shaft portion **181** of the screw **18** is inserted into the washers **191** and **193** from the outer side in parallel with the Y direction, and is screwed into the nut **194**.

(76) That is, the shaft portion **181** of the screw **18** is screwed into the nut **194** in a state in which the head portion **182** of the screw **18** abuts against the outer spacer **17** (specifically, a bottom surface of the space **173***a* of the through hole **173**) from the outer side across the washer **191** and the nut **194** screwed into the shaft portion **181** of the screw **18** abuts against the rear frame **25** (specifically, a bottom surface of the space **263***c* of the through hole **263**) from the inner side. Accordingly, the rear frame **25**, the inner spacer **16**, the upright plate **142**, and the outer spacer **17** are fastened to each other. At this time, the inner spacer **16** is arranged between the rear frame **25** and the upright plate **142**, and is in contact with each of the rear frame **25** and the upright plate **142**. Further, the spacer body **171** of the outer spacer **17** is arranged between the upright plate **142** and the head portion **182** of the screw **18**, is in contact with the upright plate **142**, and abuts on the head portion **182** of the screw **18** across the washer **191**. Furthermore, the protruding portion **172** of the outer spacer **17** is located between a peripheral edge of the through hole **144** of the upright plate **142** and the shaft portion **181** of the screw **18**.

(77) Thus, the support fitting **14** is attached to the housing **2** by the fitting attachment portion **15** so as to be rotatable about the axis Ay. Therefore, a direction in which ions are released from the static eliminator **1** can be changed by rotating the housing **2** with respect to the support fitting **14**. Further, the insulating pads **131**, **132**, **133**, and **134** are separated from the placement surface Axy in a state in which the support fitting **14** supports the housing **2** with respect to the placement surface Axy.

(78) FIG. **13** is a block diagram schematically illustrating a configuration of a controller which is the electrical equipment system of the static eliminator of FIG. **1**. The static eliminator **1** includes a controller **8** accommodated in the electrical equipment accommodating portion **202**. The controller **8** includes a fan unit controller **81** that controls the fan unit **3**, a cleaning unit controller **83** that controls the cleaning unit **7**, and an electrode unit controller **9** that controls the negative electrode unit **5** and the positive electrode unit **6**.

(79) The fan unit controller **81** rotates the fan **33** provided in the fan unit **3** to generate air flowing in the air blowing direction Dw in the fan **33**. This air flows into the housing **2** from the rear side Xb via the rear wire mesh **125**. Furthermore, after passing through the flow path Fw in the housing **2**, the air flows out from the housing **2** to the front side Xf via the front wire mesh **115** and the mesh portion **112**. The air flowing out from the housing **2** in this manner reaches the object to be

neutralized.
(80) The cleaning unit controller **83** causes the cleaning brushes **71***m* and **71***p* to clean the electrode needles Nm and Np by controlling a rotational position of the motor **72** of the cleaning unit **7**. That is, when cleaning one electrode needle Nm among the plurality of electrode needles Nm, the cleaning unit controller **83** controls the rotational position of the motor **72** to move the cleaning brush **71***m* to the cleaning position Lm opposing the one electrode needle Nm, and then, cause the cleaning brush **71***m* to slightly reciprocate in the circumferential direction Dc (a cleaning operation). Further, all of the plurality of electrode needles Nm can be cleaned by executing the cleaning operation while sequentially changing one electrode needle Nm to be cleaned among the plurality of electrode needles Nm. Similarly, when cleaning one electrode needle Np among the plurality of electrode needles Np, the cleaning unit controller **83** controls the rotational position of the motor **72** to move the cleaning brush **71***p* to the cleaning position Lp opposing the one electrode needle Np, and then, cause the cleaning brush **71***p* to slightly reciprocate in the circumferential direction Dc (a cleaning operation). Further, all of the plurality of electrode needles Np can be cleaned by executing the cleaning operation while sequentially changing one electrode needle Np to be cleaned among the plurality of electrode needles Np.
(81) As described above, the electrode unit controller **9** is connected to the negative electrode unit **5** by the harness Hm, and is connected to the positive electrode unit **6** by the harness Hp. The electrode unit controller **9** controls the voltage applied to the electrode needle Nm of the negative electrode unit **5** via the harness Hm and the voltage applied to the electrode needle Np of the positive electrode unit **6** via the harness Hp, thereby generating a corona discharge between the tip portion of the electrode needle Nm and the tip portion of the electrode needle Np. Due to this corona discharge, negative ions are generated around the tip portion of the electrode needle Nm, and positive ions are generated around the tip portion of the electrode needle Np. Furthermore, the rear wire mesh **125**, the positive electrode unit **6**, and the negative electrode unit **5** are arrayed in order in the air blowing direction Dw, and the rear wire mesh **125** is connected to the ground G. Therefore, a corona discharge is generated between the electrode needle Np and the rear wire mesh **125**, and positive ions are generated around the electrode needle Np. Similarly, a corona discharge is generated between the electrode needle Nm and the rear wire mesh **125**, and negative ions are generated around the electrode needle Nm.
(82) As described above, the electrode needle Nm and the electrode needle Np protrude to the flow path Fw, and the air generated by the fan **33** passes the tip portions of the electrode needle Nm and the electrode needle Np. Therefore, the negative ions generated around the tip portion of the electrode needle Nm and the positive ions generated around the tip portion of the electrode needle Np advance to the front side Xf with the air passing through the flow path Fw in the air blowing direction Dw. Further, the fan **33** that generates the air is located on the front side Xf of the positive electrode unit **6** and the negative electrode unit **5**, in other words, on the downstream side in the air blowing direction Dw. Therefore, the negative ions and the positive ions flow out from the housing **2** to the front side Xf via the front wire mesh **115** and the mesh portion **112** after being stirred by the fan **33**.
(83) FIG. **14** is a flowchart illustrating an example of an operation executed by the controller of FIG. **13**. In Step S**101**, the cleaning unit controller **83** starts cleaning the electrode needles Nm and the electrode needles Np. As illustrated in FIG. **8**A, in the static eliminator **1**, the electrode needles Nm and Np are alternately aligned clockwise in the circumferential direction Dc, and a total of eight electrode needles Nm and Np are aligned. On the other hand, the cleaning operations for the eight electrode needles Nm and Np are performed in order of proximity to the accommodation box **751** in the clockwise direction. More specifically, for each of the electrode needles Nm and Np, the cleaning brushes **71***m* and **71***p* are moved back and forth so as to pass the electrode needles Nm and Np, and then, the cleaning brushes **71***m* and **71***p* are moved so as to clean the next electrode needles Nm and Np. In the present embodiment, the cleaning brushes **71***m* and **71***p* are moved such that the

cleaning operation is executed for each of the electrode needles Nm and Np, but a moving method is not limited thereto. For example, it may be configured such that all the electrode needles Nm and Np are cleaned by moving the cleaning brushes **71***m* and **71***p* in one direction. Further, the cleaning operations for the electrode needles Nm and Np may be executed in order of proximity to the accommodation box **751** in the counterclockwise direction.

(84) That is, the cleaning unit controller **83** controls the rotational position of the motor **72** to move the cleaning brushes **71***m* and **71***p* from the brush cleaner **75** to the cleaning position Lm opposing the first electrode needle Nm, thereby executing the cleaning operation on this electrode needle Nm. At this time, the cleaning brushes **71***m* and **71***p* moving from the accommodation box **751** to the cleaning position Lm are slid on the sliding contact members of the brush cleaner **75**, whereby the cleaning of the cleaning brushes **71***m* and **71***p* is executed. Further, when the cleaning operation for the last (eighth) electrode needle Np is completed, the cleaning unit controller **83** controls the rotational position of the motor **72** to move the cleaning brushes **71***m* and **71***p* from the cleaning position Lp opposing the last electrode needle Np to the accommodation box **751**. At this time, the cleaning brushes **71***m* and **71***p* moving from the cleaning position Lp to the accommodation box **751** are slid on the sliding contact members of the brush cleaner **75**, whereby the cleaning of the cleaning brushes **71***m* and **71***p* is executed. Incidentally, the cleaning unit controller **83** make speeds of the cleaning brushes **71***m* and **71***p* at the time of taking out the cleaning brushes **71***m* and **71***p* from the accommodation box **751** slower than speeds of the cleaning brushes **71***m* and **71***p* at the time of putting the cleaning brushes **71***m* and **71***p* into the accommodation box **751**.

(85) In Step S**102**, the fan unit controller **81** starts rotation of the fan **33** to generate air in the air blowing direction Dw. In Step S**103**, the electrode unit controller **9** starts applying a voltage to the electrode needles Nm of the negative electrode unit **5** and applying a voltage to the electrode needles Np of the positive electrode unit **6**. As a result, a negative DC voltage Vm lower than a voltage of the ground G is applied to the electrode needles Nm, and a positive DC voltage Vp higher than the voltage of the ground G is applied to the electrode needles Np. Further, the rear wire mesh **125** is connected to the ground G. Therefore, a potential difference Vm is generated between the electrode needle Nm and the rear wire mesh **125**, a potential difference Vp is generated between the electrode needle Nm and the rear wire mesh **125**, and a potential difference Vpm (=Vp−Vm) is generated between the electrode needle Np and the electrode needle Nm. Then, negative ions and positive ions are generated by corona discharges respectively generated by the potential difference Vm, the potential difference Vp, and the potential difference Vpm. The negative ions and the positive ions thus generated advance in the air blowing direction Dw by the air and are released from the static eliminator **1** to the front side Xf (a static elimination operation). Note that the cleaning unit controller **83** controls the rotational position of the motor **72** to locate the cleaning brushes **71***m* and **71***p* in the accommodation box **751** during execution of the static elimination operation.

(86) In the voltage control in Step S**104**, feedback control for controlling ion balance in a long term and a short term is executed. Details of this voltage control will be described later with reference to FIGS. **15**A and **15**B. When the electrode unit controller **9** finishes applying the voltages to the electrode needles Nm and the electrode needles Np in Step S**105** following Step S**104**, the fan unit controller **81** stops the fan **33** and finishes air blowing performed by the fan **33** in Step S**106**.

(87) FIG. **15**A is a block diagram illustrating details of the electrode unit controller. The electrode unit controller **9** includes a central processing unit (CPU) **91**, a negative polarity high voltage power supply **92** that generates the voltage Vm to be applied to the electrode needles Nm, and a positive polarity high voltage power supply **93** that generates the voltage Vp to be applied to the electrode needles Np. The CPU **91** executes digital signal processing for controlling the negative polarity high voltage power supply **92** and the positive polarity high voltage power supply **93**. The CPU **91** includes a high voltage control unit **911** that controls the voltage Vp (high voltage) to be applied to the electrode needles Np, and a first balance control unit **912** that controls balance (ion

balance) between negative ions and positive ions generated by the application of the voltages Vp and Vm to the electrode needles Np and Nm. Specifically, the CPU **91** executes a predetermined program to configure the high voltage control unit **911** and the first balance control unit **912**.
(88) The negative polarity high voltage power supply **92** is a transformer having a primary-side circuit **921** and a secondary-side circuit **922**. A voltage signal Vim is input to the primary-side circuit **921**, and the secondary-side circuit **922** is connected to each of the electrode needles Nm of the negative electrode unit **5** by the harness Hm. Then, the voltage Vm corresponding to the voltage signal Vim input to the primary-side circuit **921** is applied to each of the electrode needles Nm from the secondary-side circuit **922** via the harness Hm.
(89) The positive polarity high voltage power supply **93** is a transformer having a primary-side circuit **931** and a secondary-side circuit **932**. A voltage signal Vip is input to the primary-side circuit **931**, and the secondary-side circuit **932** is connected to each of the electrode needles Np of the positive electrode unit **6** by the harness Hp. Then, the voltage Vp corresponding to the voltage signal Vip input to the primary-side circuit **931** is applied to each of the electrode needles Np from the secondary-side circuit **932** via the harness Hp.
(90) In the housing **2**, the above-described ground G (internal ground) is provided. The rear frame **25** made of the antistatic resin in the housing **2** is short-circuited to the ground G. Note that a mode of electrically connecting the rear frame **25** and the ground G is not limited to the short circuit, and these may be connected via a resistor.
(91) Further, the electrode unit controller **9** includes: a ground electrode Te short-circuited to an earth E (external ground); and a low-response detection circuit **94** provided between the ground electrode Te and the ground G. The low-response detection circuit **94** includes a detection resistor R**94** that connects the ground electrode Te and the ground G. The detection resistor R**94** is provided to detect a current Idl flowing into the static eliminator **1** from the earth E via the ground electrode Te. That is, when there is a difference between the amount of the negative ions and the amount of the positive ions released from the static eliminator **1**, an electric charge corresponding to the difference flows from the earth E into the ground electrode Te, and the current Idl due to the electric charge flows to the detection resistor R**94**. As a result, a voltage Vdl corresponding to the current Idl is generated at a detection point **941** between the detection resistor R**94** and the ground G. In this manner, the low-response detection circuit **94** converts the current Idl, generated by the electric charge flowing into the housing **2** from the earth E via the ground electrode Te, into the voltage Vdl by the detection resistor R**94**. In other words, the low-response detection circuit **94** detects the voltage Vdl indicating ion balance between the negative ions and the positive ions generated by the static eliminator **1** and absorbed by the earth E.
(92) Furthermore, the electrode unit controller **9** includes a high-response detection circuit **95** provided between the front wire mesh **115** and the ground G. The high-response detection circuit **95** includes a detection resistor R**95** that connects the front wire mesh **115** and the ground G. The detection resistor R**95** is provided to detect a current Idh flowing from the front wire mesh **115** to the ground G. That is, the negative ions and the positive ions generated around the electrode needle Nm and the electrode needle Np move in the air blowing direction Dw and arrive at the front wire mesh **115**. The negative ions and the positive ions that have arrived at the front wire mesh **115** in this manner are partially absorbed by the front wire mesh **115**. Therefore, an electric charge corresponding to a difference between the amount of the negative ions and the amount of the positive ions absorbed by the front wire mesh **115** flows from the front wire mesh **115** toward the ground G, and the current Idh due to this electric charge flows to the detection resistor R**95**. As a result, a voltage Vdh corresponding to the current Idh is generated at a detection point **951** between the detection resistor R**95** and the front wire mesh **115**. In this manner, the high-response detection circuit **95** converts the current Idh, generated by the electric charge flowing from the front wire mesh **115** to the ground G, into the voltage Vdh by the detection resistor R**95**. In other words, the high-response detection circuit **95** detects the voltage Vdh indicating ion balance between the

negative ions and the positive ions generated by the static eliminator **1** and absorbed by the front wire mesh **115**.
(93) Here, a resistance value of the detection resistor R**94** of the low-response detection circuit **94** is larger than a resistance value of the detection resistor R**95** of the high-response detection circuit **95**. Further, the capacitance of the earth E is larger than the capacitance of the front wire mesh **115**. Therefore, a time constant of the high-response detection circuit **95** is lower than a time constant of the low-response detection circuit **94**, in other words, a response speed of the high-response detection circuit **95** is faster than a response speed of the low-response detection circuit **94**. That is, the high-response detection circuit **95** detects a fluctuation of a high frequency out of fluctuations of the ion balance, and the low-response detection circuit **94** detects a fluctuation of a low frequency lower than the high frequency out of the fluctuations of the ion balance.
(94) The electrode unit controller **9** controls the ion balance by executing feedback control on the voltages Vm and Vp to be applied to the electrode needles Nm and Np based on the fluctuations of the ion balance detected by the low-response detection circuit **94** and the high-response detection circuit **95**. Specifically, the electrode unit controller **9** includes a second balance control unit **96** that suppresses fluctuations (wobble) of the ion balance, and the feedback control is executed by the second balance control unit **96**.
(95) More specifically, the low-response detection circuit **94** outputs the voltage Vdl indicating the fluctuation of the ion balance in the low frequency to the first balance control unit **912** of the CPU **91**. The first balance control unit **912** holds a target voltage Vtl which is a target value of the voltage Vdl, generates a voltage signal Vs according to a difference between the voltage Vdl and the target voltage Vtl, and outputs the voltage signal Vs to the second balance control unit **96**. Incidentally, the target voltage Vtl is set to zero volt. That is, a target state is a state in which the amount of the negative ions and the amount of the positive ions released from the static eliminator **1** become equal to each other and the electric charge flowing from the earth E into the static eliminator **1** becomes zero.
(96) Further, the high-response detection circuit **95** outputs the voltage Vdh indicating the fluctuation of the ion balance in the high frequency to the second balance control unit **96**. In regard to this, the second balance control unit **96** holds a target voltage Vth which is a target value of the voltage Vdh, generates the voltage signal Vim, which is a control signal for performing the feedback control of the voltage Vm according to a difference between the voltage Vdh and the target voltage Vth and the voltage signal Vs, and outputs the voltage signal Vim to the primary-side circuit **921** of the negative polarity high voltage power supply **92**. Incidentally, the target voltage Vth is set to not zero volt but a voltage shifted from zero by a predetermined offset voltage. That is, there is a difference between the ease of absorption of the negative ions by the front wire mesh **115** and the ease of absorption of the positive ions by the front wire mesh **115**. Therefore, in a target state in which the equal amounts of the negative ions and the positive ions arrive at the front wire mesh **115**, the current Idh does not become zero, and the voltage Vdh is shifted by an offset voltage Vo (offset amount) from the voltage (zero volt) of the ground G. Therefore, the target voltage Vth of the voltage Vdh is set to the offset voltage Vo. Note that the offset voltage Vo is experimentally measured in advance and set in the second balance control unit **96**.
(97) In this manner, the feedback control for converging the voltage Vdl toward the target voltage Vtl and the feedback control for converging the voltage Vdh toward the target voltage Vth are executed. In other words, the feedback control for converging the current Idl to a target current Itl (=Vtl/R**97**) and the feedback control for converging the current Idh to a target current Ith (=Vth/R**95**) are executed. Note that the second balance control unit **96** that executes such control may be configured using an analog circuit such as an operational amplifier or may be configured using a digital circuit such as a processor.
(98) Further, the electrode unit controller **9** executes control for applying voltages Vp and Vm, which are necessary and sufficient for the electrode needles Np and Nm to generate the corona

discharges, to the electrode needles Np and Nm using the rear wire mesh **125**. More specifically, since the rear wire mesh **125** is short-circuited to the ground G, the electric charge generated in the rear wire mesh **125** flows from the rear wire mesh **125** to the ground G. Note that a mode of electrically connecting the rear wire mesh **125** and the ground G is not limited to the short circuit, and these may be connected via a resistor.

(99) Specifically, along a circuit formed by a corona discharge between the electrode needle Nm and the rear wire mesh **125**, a current Irn corresponding to an electric charge generated by the corona discharge flows from the rear wire mesh **125** to the ground G. Further, along a circuit formed by a corona discharge between the electrode needle Np and the rear wire mesh **125**, a current Irp corresponding to an electric charge generated by the corona discharge flows from the rear wire mesh **125** to the ground G. On the other hand, the secondary-side circuit **922** of the negative polarity high voltage power supply **92** is connected to the ground G, and the secondary-side circuit **932** of the positive polarity high voltage power supply **93** is connected to the ground G. Therefore, a current Ign mainly including the current Irn reaching the ground G from the rear wire mesh **125** flows from the ground G to the secondary-side circuit **922**, and a current Igp mainly including the current Irp reaching the ground G from the rear wire mesh **125** flows from the ground G to the secondary-side circuit **932**.

(100) Further, the electrode unit controller **9** also includes a discharge amount detection circuit **97** provided between the secondary-side circuit **932** of the positive polarity high voltage power supply **93** and the ground G. The discharge amount detection circuit **97** includes a detection resistor R**97** that connects the secondary-side circuit **932** and the ground G. Therefore, the current Igp flowing from the ground G to the secondary-side circuit **932** flows through the detection resistor R**97**. As a result, a voltage Vgp corresponding to the current Igp is generated at a detection point **971** between the detection resistor R**97** and the secondary-side circuit **932**. As described above, the discharge amount detection circuit **97** converts the current Igp, which flows from the rear wire mesh **125** to the secondary-side circuit **932** of the positive polarity high voltage power supply **93** via the ground G, into the voltage Vgp by the detection resistor R**97**. In other words, the discharge amount detection circuit **97** detects the voltage Vgp indicating an amount of positive ions generated in response to the application of the voltage Vp to the electrode needle Np.

(101) The discharge amount detection circuit **97** outputs the detected voltage Vgp to the high voltage control unit **911** of the CPU **91**. The high voltage control unit **911** holds a target voltage Vtp which is a target value of the voltage Vgp, generates the voltage signal Vip which is a control signal for performing the feedback control of the voltage Vp according to a difference between the voltage Vgp and the target voltage Vtp, and outputs the voltage signal Vip to the primary-side circuit **931** of the positive polarity high voltage power supply **93**. As a result, the feedback control for converging the voltage Vgp toward the target voltage Vtp is executed. As a result, the positive ions in the amount corresponding to the target voltage Vtp are generated around the electrode needle Np. Note that, as described above, the second balance control unit **96** or the like also executes the feedback control to balance the generation amount of negative ions and the generation amount of the positive ions. Therefore, the negative ions are generated around the electrode needle Nm so as to follow the positive ions generated around the electrode needle Np. As a result, the negative ions in the amount corresponding to the target voltage Vtp are generated around the electrode needle Nm. Such control increases the voltages to be applied to the electrode needles Nm and Np in accordance with the progress of wear of the electrode needles Nm and Np, and the amount of negative ions and the amount of positive ions generated in accordance with the corona discharges by the electrode needles Nm and Np are maintained constant.

(102) FIG. **15**B is a flowchart illustrating an example of the voltage control executed in the operation of FIG. **14**. In Step S**201**, the target voltage Vtl for controlling the ion balance in the long term and the target voltage Vth for controlling the ion balance in the short term are acquired by the first balance control unit **912** and the second balance control unit **96**. Then, the voltage Vdl

detected by the low-response detection circuit **94** is acquired by the first balance control unit **912** in Step S**202**, and the voltage Vdh detected by the high-response detection circuit **95** is acquired by the second balance control unit **96** in Step S**203**. Then, in a case where the voltage Vdl has changed by a certain amount ("YES" in Step S**204**), the second balance control unit **96** executes the feedback control based on the target voltage Vtl and the voltage Vdl and the feedback control based on the target voltage Vth and the voltage Vdh, and inputs the voltage signal Vim to the negative polarity high voltage power supply **92** (Step S**205**). On the other hand, in a case where the voltage Vdl has not changed by the certain amount ("NO" in Step S**204**), the second balance control unit **96** executes the feedback control based on the target voltage Vth and the voltage Vdh, and inputs the voltage signal Vim to the negative polarity high voltage power supply **92** (Step S**206**).

(103) In the static eliminator **1** described above, the electrode needles Np and Nm (an ion generator) that generate the positive ions and negative ions, and the positive polarity high voltage power supply **93** and the negative polarity high voltage power supply **92** (a high voltage application unit) that apply the voltage Vp (a positive polarity high voltage) and the voltage Vm (a negative polarity high voltage) to the electrode needles Np and Nm are provided. Then, the electrode needle Np generates the positive ions when the positive polarity high voltage power supply **93** applies the voltage Vp to the electrode needle Np, and the electrode needle Nm generates the negative ions when the negative polarity high voltage power supply **92** applies the voltage Vm to the electrode needle Nm. Further, the current Idl (an ion current) flowing between the earth E and the static eliminator **1** via the ground electrode Te is detected, and the feedback control is executed on the negative polarity high voltage power supply **92** such that the current Idl becomes the target current Itl. The ion balance can be appropriately controlled by the feedback control based on the current Idl. Further, the housing **2** is provided with the conductive rear frame **25** (conductive member) in order to suppress charging of the housing **2** of the static eliminator **1**. The rear frame **25** is insulated from the placement surface Axy of the static eliminator **1** by the insulators, such as the insulating pads **131**, **132**, **133**, and **134**, the inner spacer **16**, and the outer spacer **17**, so that an electric charge can be prevented from moving from the rear frame **25** to the earth E via the placement surface Axy. In addition, the rear frame **25** is not connected to the earth E, but is connected to a wire electrically connected to each of the low-response detection circuit **94** (a detection circuit), the positive polarity high voltage power supply **93**, and the negative polarity high voltage power supply **92**, that is, the ground G. As a result, the electric charge of the rear frame **25** is absorbed by the positive polarity high voltage power supply **93** and the negative polarity high voltage power supply **92**, so that the electric charge can be prevented from moving from the rear frame **25** to the earth E. As a result, it is possible to suppress the charging of the housing **2** of the static eliminator **1** while avoiding an influence on the control of the ion balance. Incidentally, the ground G can be configured using a wire made of metal such as copper.

(104) Further, the insulating pads **131**, **132**, **133**, and **134** (a support member), which have insulating properties and are attached to the rear frame **25** on the bottom surface **2**B of the housing **2** are provided. In a state in which the housing **2** is placed on the placement surface Axy, the insulating pads **131**, **132**, **133**, and **134** are in contact with the placement surface Axy between the rear frame **25** and the placement surface Axy to separate the rear frame **25** from the placement surface Axy. In such a configuration, the rear frame **25** and the placement surface Axy are separated from each other by the insulating pads **131**, **132**, **133**, and **134** each of which has the insulating property and is attached to the rear frame **25** on the bottom surface **2**B of the housing **2**, so that the electric charge can be prevented from moving from the rear frame **25** to the earth E via the placement surface Axy.

(105) Further, the support fitting **14** made of metal and the fitting attachment portion **15** that rotatably supports the support fitting **14** with respect to the rear frame **25** on the side surface of the housing **2** are provided. The support fitting **14** comes into contact with the placement surface Axy and supports the housing **2** with respect to the placement surface Axy, thereby separating the

housing **2** from the placement surface Axy. Further, the fitting attachment portion **15** includes the insulating inner spacer **16** (first spacer) that is arranged between the rear frame **25** and the support fitting **14** on the side surface of the housing **2** and restricts the contact between the rear frame **25** and the support fitting **14**. In such a configuration, the contact between the rear frame **25** and the support fitting **14** in contact with the placement surface Axy is restricted by the inner spacer **16**, so that the electric charge can be prevented from moving from the rear frame **25** to the earth E via the support fitting **14** and the placement surface Axy.

(106) Further, the fitting attachment portion **15** further includes the insulating outer spacer **17** (second spacer) abutting on the support fitting **14** from the opposite side (outer side) of the inner spacer **16**, and the screw **18** made of metal. The through holes **161**, **144**, and **173** (insertion holes) into which the shaft portion **181** of the screw **18** is inserted are opened in the inner spacer **16**, the support fitting **14**, and the outer spacer **17**, respectively, and the inner spacer **16**, the support fitting **14**, and the outer spacer **17** are fastened to the housing **2** by the screw **18** in the state of being sandwiched between the head portion **182** of the screw **18** and the rear frame **25**. On the other hand, the outer spacer **17** has the spacer body **171** and the protruding portion **172** protruding from the spacer body **171** to the inner spacer **16** side (inner side). The spacer body **171** is located between the head portion **182** of the screw **18** and the support fitting **14** to restrict the contact between the head portion **182** of the screw **18** and the support fitting **14**, and the protruding portion **172** is located between the peripheral edge of the through hole **144** provided in the support fitting **14** and the shaft portion **181** of the screw **18** to restrict the contact between the support fitting **14** and the shaft portion **181** of the screw **18**. In such a configuration, the contact between the metal screw **18** for fastening the support fitting **14** to the housing **2** and the support fitting **14** is restricted by the outer spacer **17**. Therefore, even if the rear frame **25** and the screw **18** come into contact with each other, the electric charge can be prevented from moving from the rear frame **25** to the support fitting **14** via the screw **18**, and eventually, it is possible to prevent the movement of the electric charge from the rear frame **25** to the earth E.

(107) As described above, in the present embodiment, the static eliminator **1** corresponds to an example of a "static eliminator" of the invention; the insulating pads **131**, **132**, **133**, and **134** correspond to examples of a "support member" of the invention; the support fitting **14** corresponds to an example of a "support fitting" of the invention; the fitting attachment portion **15** corresponds to an example of a "fitting attachment portion" of the invention; the inner spacer **16** corresponds to an example of a "first spacer" of the invention; the through holes **161**, **144**, and **173** correspond to examples of "insertion holes" of the invention; the outer spacer **17** corresponds to an example of a "second spacer" of the invention; the spacer body **171** corresponds to an example of a "spacer body" of the invention; the protruding portion **172** corresponds to an example of a "protruding portion" of the invention; the screw **18** corresponds to an example of a "screw" of the invention; the shaft portion **181** corresponds to an example of a "shaft portion" of the invention; the head portion **182** corresponds to an example of a "head portion" of the invention; the housing **2** corresponds to an example of a "housing" of the invention; the rear frame **25** corresponds to an example of a "conductive member" of the invention; the positive polarity high voltage power supply **93** and the negative polarity high voltage power supply **92** cooperate to function as an example of a "high voltage application unit" of the invention; the low-response detection circuit **94** corresponds to an example of a "detection circuit" of the invention; the second balance control unit **96** corresponds to an example of a "feedback control unit" of the invention; the earth E corresponds to an example of an "earth" of the invention; the ground G corresponds to an example of a "wire" of the invention; the current Idl corresponds to an example of an "ion current" of the invention; the target current Ith corresponds to an example of a "target value" of the invention; the electrode needles Np and Nm correspond to examples of an "ion generator" of the invention; the ground electrode Te corresponds to an example of a "ground electrode" of the invention; the voltage Vp corresponds to an example of a "positive polarity high voltage" of the invention; and the voltage

Vm corresponds to an example of a “negative polarity high voltage” of the invention.
(108) Note that the invention is not limited to the above-described embodiments and various modifications can be made to those described above without departing from the gist thereof. For example, the specific configuration of the conductive member that imparts conductivity to the housing **2** is not limited to the antistatic member, and may be metal or a conductive resin.
(109) Further, in the housing **2**, the conductivity may be imparted to a member other than the rear frame **25**, for example, the front frame **21**. In this case, the rear frame **25** may be an insulator.
(110) Further, the first unit frame **51** and the second unit frame **61** do not need to have an arc shape, and may have a circular shape.
(111) Further, an arrangement mode of the electrode needles Nm and Np in the first and second unit frames **51** and **61** may be changed. For example, the electrode needles Nm and Np may be provided so as to protrude outward from the outer walls **512** and **612** of the first and second unit frames **51** and **61**.
(112) Further, the number or the arrangement mode of the electrode needles Nm and Np may be appropriately changed.
(113) Further, an arrangement order of the negative electrode unit **5** and the positive electrode unit **6** in the X direction may be reversed.
(114) Further, the fan unit **3** may be arranged on the upstream side of the negative electrode unit **5** and the positive electrode unit **6** in the air blowing direction Dw.
(115) Further, specific contents of the control of a generation amount of ions executed by the high voltage control unit **911** are not limited to the above example. That is, the generation amount of ions may be controlled by performing the feedback control on the voltage Vm based on the current Ign flowing from the ground G to the secondary-side circuit **922** of the negative polarity high voltage power supply **92**.
(116) Further, the control for generating the predetermined amount of ions regardless of the progress of wear of the electrode needles Nm and Np (control by the high voltage control unit **911**) is executed on the positive polarity high voltage power supply **93**, and the control for realizing appropriate ion balance (control by the second balance control unit **96**) is executed on the negative polarity high voltage power supply **92**. However, the former control may be executed on the negative polarity high voltage power supply **92**, and the latter control may be executed on the positive polarity high voltage power supply **93**.
(117) Further, two types of the electrode needles Np and Nm to which different DC voltages Vp and Vm are applied are provided, and the positive ions are generated by the electrode needle Np, and the negative ions are generated by the electrode needle Nm. However, positive ions and negative ions may be generated by corona discharges generated by applying an AC voltage, which varies with time between the voltage Vp and the voltage Vm, to one type of electrode needle.
(118) Further, the negative electrode unit **5** and the positive electrode unit **6** may be configured as illustrated in FIG. **16**. FIG. **16** is a perspective view schematically illustrating a modified example of the negative electrode unit and the positive electrode unit. In the modified example illustrated in FIG. **16**, the negative electrode unit **5** includes the first unit frame **51** having a flat plate shape extending in the Y direction, and the plurality electrode needles Nm are arrayed in the Y direction on a rear end surface of the first unit frame **51**. Each of the electrode needles Nm protrudes from the rear end surface of the first unit frame **51** to the rear side Xb in the X direction. Further, the positive electrode unit **6** includes the second unit frame **61** having a flat plate shape extending in the Y direction, and the plurality of electrode needles Np are arrayed in the Y direction on a rear end surface of the second unit frame **61**. Each of the electrode needles Np protrudes from the rear end surface of the second unit frame **61** to the rear side Xb in the X direction. The electrode needle Nm and the electrode needle Np generate negative ions and positive ions in response to application of a voltage. The negative ions and the positive ions are released from the static eliminator **1** by air in the air blowing direction Dw parallel to the X direction.

(119) Further, the static eliminator **1** described above is provided with the system performing the feedback control of ion balance in the long term and the system performing the feedback control of ion balance in the short term. A specific configuration for executing such two feedback control systems is not limited to the example of FIG. **15**A. That is, any configuration that realizes the two feedback systems conceptually illustrated in FIG. **17** can be adopted.
(120) FIG. **17** is a diagram schematically illustrating two systems that perform long-term feedback and short-term feedback. Positive ions and negative ions whose ion balance is controlled by ion output control **981** are emitted from the housing **2** to an external target space via the front cover **11**. Then, first ion balance **982** indicating the ion balance in the target space is detected, and the first ion balance **982** is fed back to the ion output control **981** by a feedback loop **983**. The ion output control **981** executes long-term feedback control (that is, feedback control with a low response speed) for bringing the first ion balance **982** closer to a target value on the ion balance released from the ion output control **981**.
(121) Further, second ion balance **984** indicating ion balance at a position (for example, the inner side of the front cover **11**) different from the first ion balance **982** is detected, and the second ion balance **984** is fed back to the ion output control **981** by a feedback loop **985**. The ion output control **981** executes short-term feedback control (that is, feedback control with a high response speed) based on the second ion balance **984** on the ion balance released from the ion output control **981**.
(122) That is, first feedback control based on the first ion balance **982** and second feedback control based on the second ion balance **984** are executed, and the responsiveness of the second feedback control is higher than the responsiveness of the first feedback control. As a result, the ion balance can be appropriately maintained in the long term and the short term.
(123) Further, an ion balance sensor illustrated in FIG. **18** may be used to execute long-term feedback control. FIG. **18** is a perspective view illustrating an example of the ion balance sensor. An ion balance sensor **99** of FIG. **18** includes a sensor plate **991** that detects ion balance and an output terminal **992** that outputs a current (first ion current) according to the ion balance detected by the sensor plate **991**. At least the sensor plate **991** of the ion balance sensor **99** is arranged at an external detection position outside a device body of the static eliminator **1** including the housing **2** and the front cover **11**. Then, the ion balance (that is, the first ion balance **982**) at the external detection position is detected by the sensor plate **991**, and the first ion current is output from the output terminal **992**. The first ion current output from the output terminal **992** is fed back to the ion output control **981** by the feedback loop **983**.
(124) Note that, in a case where the ion balance sensor **99** is used for the electrode unit controller **9** in FIG. **15**A, the first ion current output from the output terminal **992** of the ion balance sensor **99** is input to, for example, a detection resistor provided in parallel with the detection resistor R**94**, and the first ion current is converted into a voltage by the detection resistor. Then, feedback control is executed by the first balance control unit **912** and the second balance control unit **96** such that the voltage corresponding to the first ion current becomes a predetermined target voltage (in other words, the first ion current becomes a predetermined target current). Note that the voltage Vdl obtained by converting the current Idl from the earth E is not reflected in the feedback control and is ignored. That is, the first balance control unit **912** and the second balance control unit **96** execute the long-term feedback control based on the first ion current detected by the ion balance sensor **99**, instead of the current Idl from the earth E.
(125) The invention is applicable to all the techniques for releasing ions generated by applying a voltage to an electrode to an object to eliminate static electricity of the object.

Claims

1. A static eliminator that releases ions to an object to eliminate static electricity of the object, the static eliminator comprising: an ion generator that generates a corona discharge in response to application of a positive polarity high voltage to generate positive ions, and generates a corona discharge in response to application of a negative polarity high voltage to generate negative ions; a high voltage application unit that applies the positive polarity high voltage and the negative polarity high voltage to the ion generator; a ground electrode short-circuited to an earth; a detection circuit that detects an ion current flowing between the earth and the static eliminator via the ground electrode; a feedback control unit that executes feedback control on the high voltage application unit to make the ion current detected by the detection circuit be a predetermined target value; a wire electrically connected to each of the detection circuit and the high voltage application unit; and a housing having a conductive member, which is insulated from a placement surface on which the static eliminator is placed and electrically connected to the wire, the housing accommodating the detection circuit.
2. The static eliminator according to claim 1, further comprising a support member that has an insulating property and is attached to the conductive member on a bottom surface of the housing, wherein, in a state in which the static eliminator is placed on the placement surface, the support member is in contact with the placement surface between the conductive member and the placement surface to separate the conductive member from the placement surface.
3. The static eliminator according to claim 1, further comprising: a support fitting made of metal; and a fitting attachment portion that rotatably supports the support fitting with respect to the conductive member on a side surface of the housing, wherein the support fitting comes into contact with the placement surface and supports the housing with respect to the placement surface to separate the housing from the placement surface, and the fitting attachment portion includes a first spacer that has an insulating property, is arranged between the conductive member and the support fitting on the side surface of the housing, and restricts contact between the conductive member and the support fitting.
4. The static eliminator according to claim 3, wherein the fitting attachment portion further includes: a second spacer that has an insulating property and abuts on the support fitting from an opposite side of the first spacer; and a screw made of metal, and an insertion hole into which a shaft portion of the screw is inserted is opened in each of the first spacer, the support fitting, and the second spacer, the first spacer, the support fitting, and the second spacer are fastened to the housing by the screw in a state of being sandwiched between a head portion of the screw and the conductive member, the second spacer includes a spacer body and a protruding portion protruding from the spacer body toward the first spacer, the spacer body is located between the head portion and the support fitting to restrict contact between the head portion and the support fitting, and the protruding portion is located between a peripheral edge of the insertion hole provided in the support fitting and the shaft portion to restrict contact between the support fitting and the shaft portion.
5. The static eliminator according to claim 1, wherein the conductive member is an antistatic member made of an antistatic resin.
6. The static eliminator according to claim 1, further comprising: a fan that causes the ions to be released from the static eliminator, the ions being generated from the ion generator; and a front wire mesh electrically connected to the wire and located a downstream side of the fan in a flow path formed by the fan.
7. The static eliminator according to claim 6, wherein the front wire mesh is electrically connected to the wire via a detection resistor.
8. The static eliminator according to claim 1, further comprising: a fan that causes the ions to be released from the static eliminator, the ions being generated from the ion generator; and a rear wire mesh electrically connected to the wire and located an upstream side of the fan in a flow path

formed by the fan.
9. The static eliminator according to claim 8, wherein a corona discharge is generated between the rear wire mesh and the ion generator.
10. The static eliminator according to claim 1, further comprising a fan that causes the ions to be released from the static eliminator, the ions being generated from the ion generator wherein the housing includes a cover frame that guides air blown by the fan and is made of an antistatic resin.
11. The static eliminator according to claim 1, wherein the wire is a ground of the high voltage application unit.